

Module 14: Macroevolution - Practice Quiz

Use this low-stakes quiz after reviewing the module keys and learning questions.

1. Under the biological species concept, two groups count as different species when they:
2. A. look slightly different from each other
3. B. simply live in different places
4. C. cannot interbreed to produce fertile offspring
5. D. have any genetic difference at all
6. Answer: C

7. Why: The biological species concept centers on reproductive isolation: no successful gene flow between the groups.

8. Two related species breed at different times of year and so never mate. This is an example of a:
9. A. prezygotic barrier
10. B. postzygotic barrier
11. C. fossil record gap
12. D. new mutation
13. Answer: A

14. Why: Prezygotic barriers act before fertilization (timing, behavior, habitat). Postzygotic barriers act after a zygote forms.

15. A mountain range splits one population into two groups that slowly diverge into separate species. This is:
16. A. sympatric speciation
17. B. no speciation at all
18. C. simple hybridization
19. D. allopatric speciation
20. Answer: D

21. Why: Allopatric speciation involves geographic separation that reduces gene flow. Sympatric speciation happens without such separation.
22. A mule, the offspring of a horse and a donkey, is healthy but sterile. This illustrates a:
23. A. prezygotic barrier
24. B. postzygotic barrier
25. C. complete lack of any isolation
26. D. brand-new species forming instantly
27. Answer: B
28. Why: Hybrid sterility is a postzygotic barrier: it acts after the zygote forms and still blocks gene flow between the parent species.